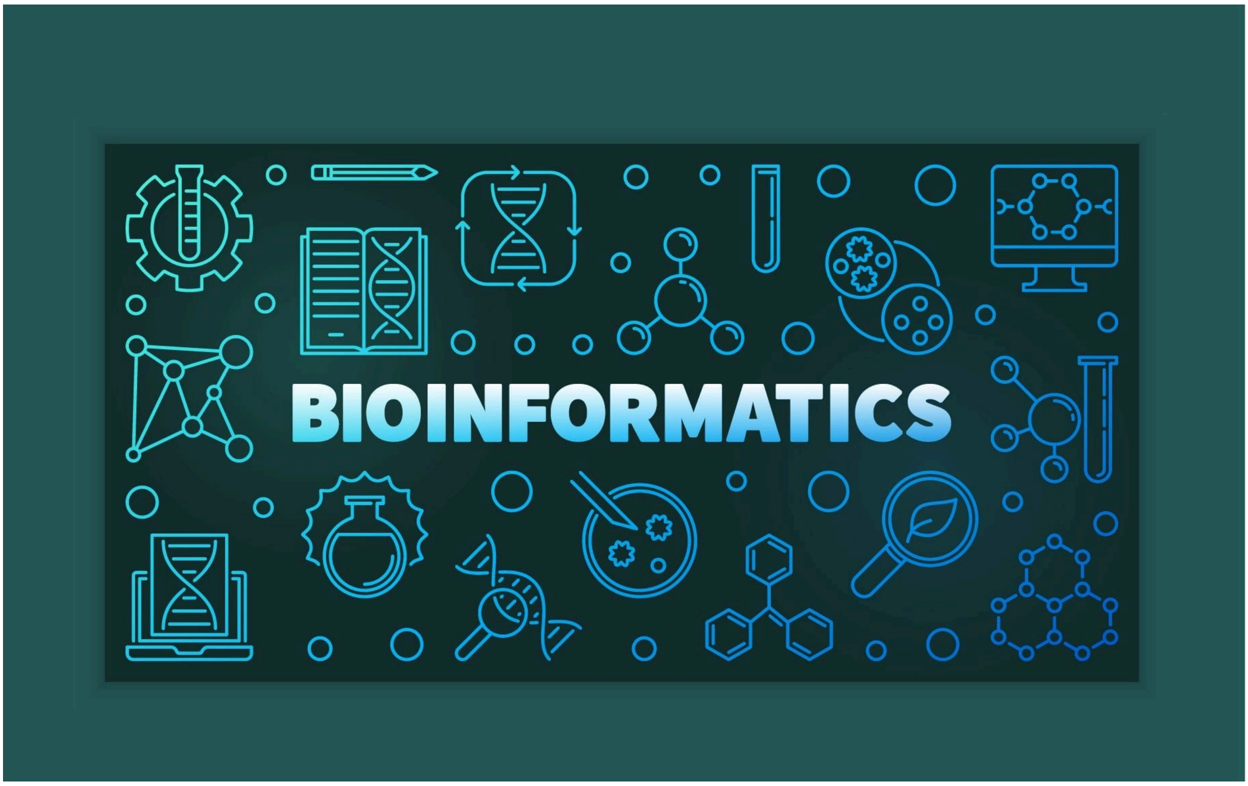
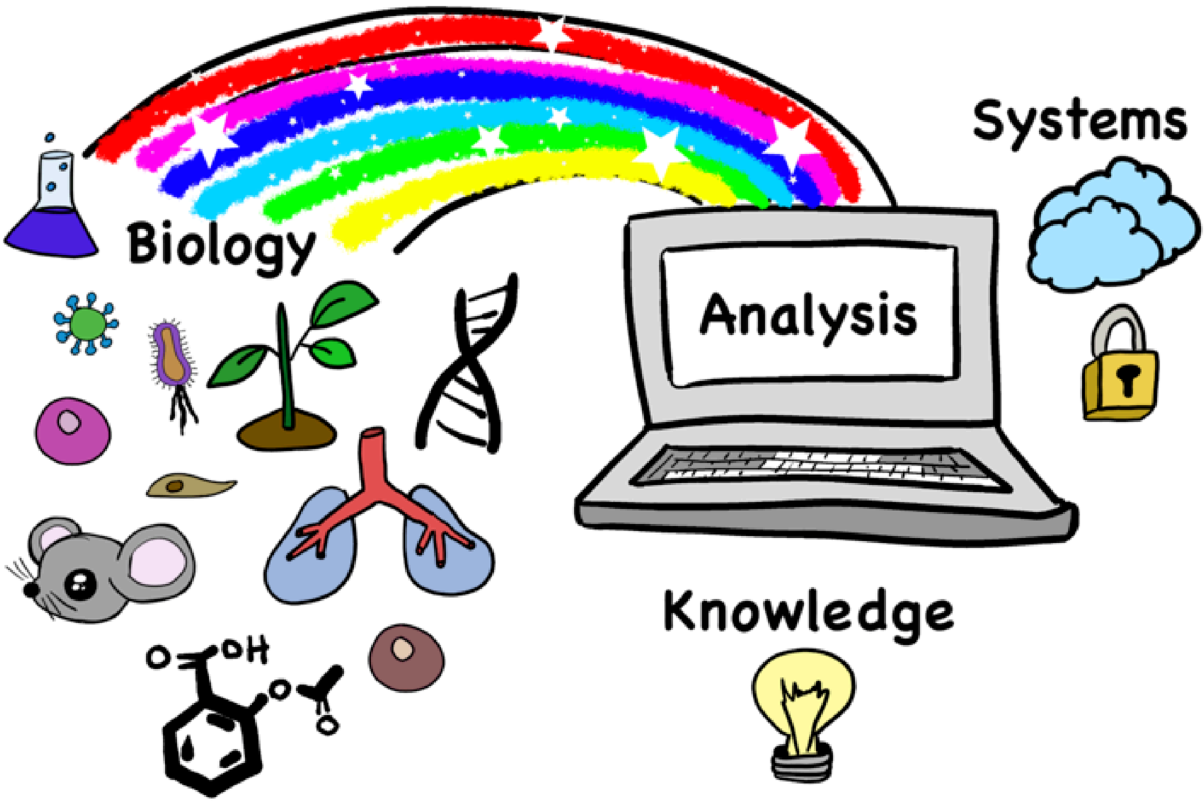


Introducció a la Bioinformàtica



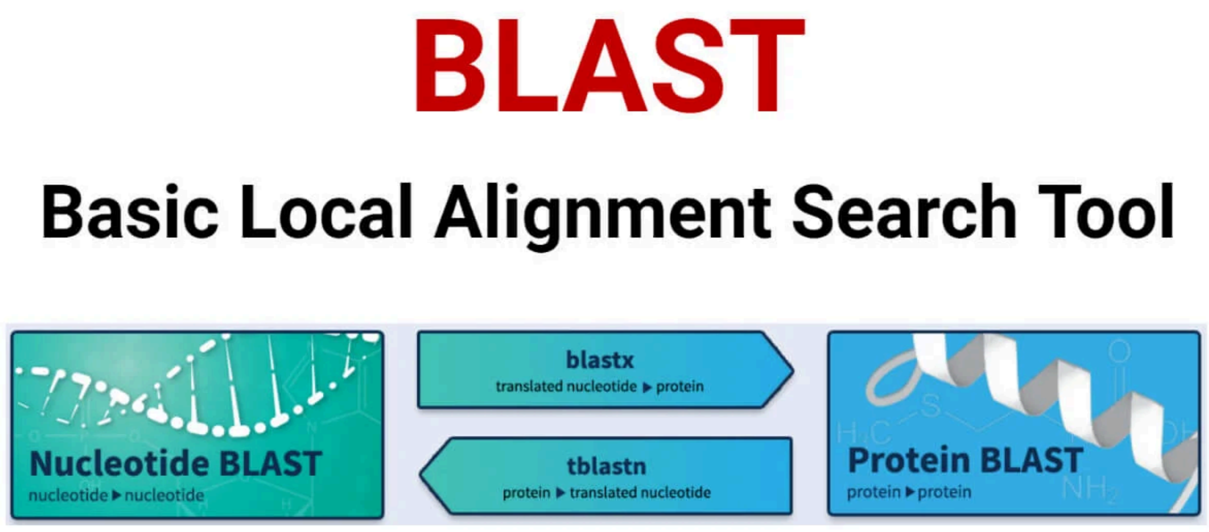
🌐INTRODUCCIÓ A LA BIOINFORMÀTICA (8h)

- Introducció (dia 1)
- Bases de dades (dia 1)
- Alineament de seqüències (dia1 / dia2)

- **Cerques per Similitud (dia 2 / dia 3)**
- Next Generation Sequenting (NGS) (dia 3 / dia 4)
- Gene Expression Omnibus database (GEO) (dia 4)

Cerca per Similitud

En aquesta sessió tractarem el tema de com fer cerques per similitud de seqüència en bases de dades que contenen múltiples seqüències (poden ser centenars de milers en bases públiques).



El procés bàsic consisteix a cercar quines seqüències d'una base de dades tenen una similitud per sobre d'un **llindar determinat** amb la nostra seqüència.

- **Seqüència problema:** La que nosaltres proporcionem (Query).
- **Objectiu:** Seleccionar aquelles que tinguin suficient similitud.

Similitud vs Homologia

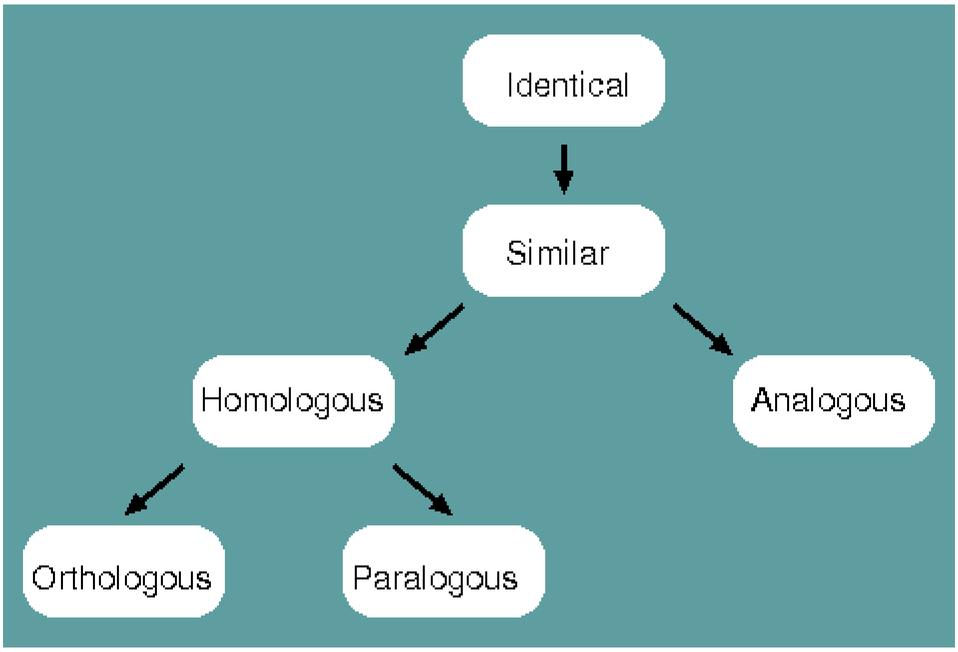
Quan obtenim una llista de seqüències ordenada per puntuació (score), estem veient **similitud**. L'investigador ha de comprovar quines són realment **homòlogues**.

Zones d'Homologia (Identitat %)

Podem classificar la confiança en l'homologia segons el percentatge d'identitat:

1. **Midnight Zone (< 20%):** No podem saber amb confiança si hi ha homologia; pot ser soroll de fons o divergència extrema.
2. **Twilight Zone (20% - 40%):** Zona ambigua. Calen anàlisis posteriors per decidir.

3. **Safe Zone (> 40%):** És força factible que siguin homòlogues, amb estructures i funcions similars.



- Identical - When a corresponding character is shared between two species or populations, that character is said to be identical.
- Similar - The degree to which two species or populations share identities.
- Homologous - When characters are similar due to common ancestry, they are homologous.
- Analogous - When characters are similar due to convergent evolution, they are analogous.
- Orthologous - When characters are homologous with conserverd function, they are orthologous.
- Paralogous - When characters are homologous with divergent function, they are paralogous.

Homology is therefore NOT synonomous with similarity. Homology is a judgement, similarity is a measurement.

Aproximacions Algorísmiques

Per determinar la similitud, hem de comparar la nostra seqüència amb totes les de la base de dades.

1. Algoritme Òptim (Smith-Waterman)

Equival a fer alineaments locals òptims per a totes les parelles.

- **Avantatge:** Màxima puntuació possible.
- **Inconvenient:** Molt lent. La complexitat computacional és:

$$O(N \times M)$$

(On N i M són les longituds de les seqüències comparades).

2. Mètodes Heurístics (FASTA, BLAST)

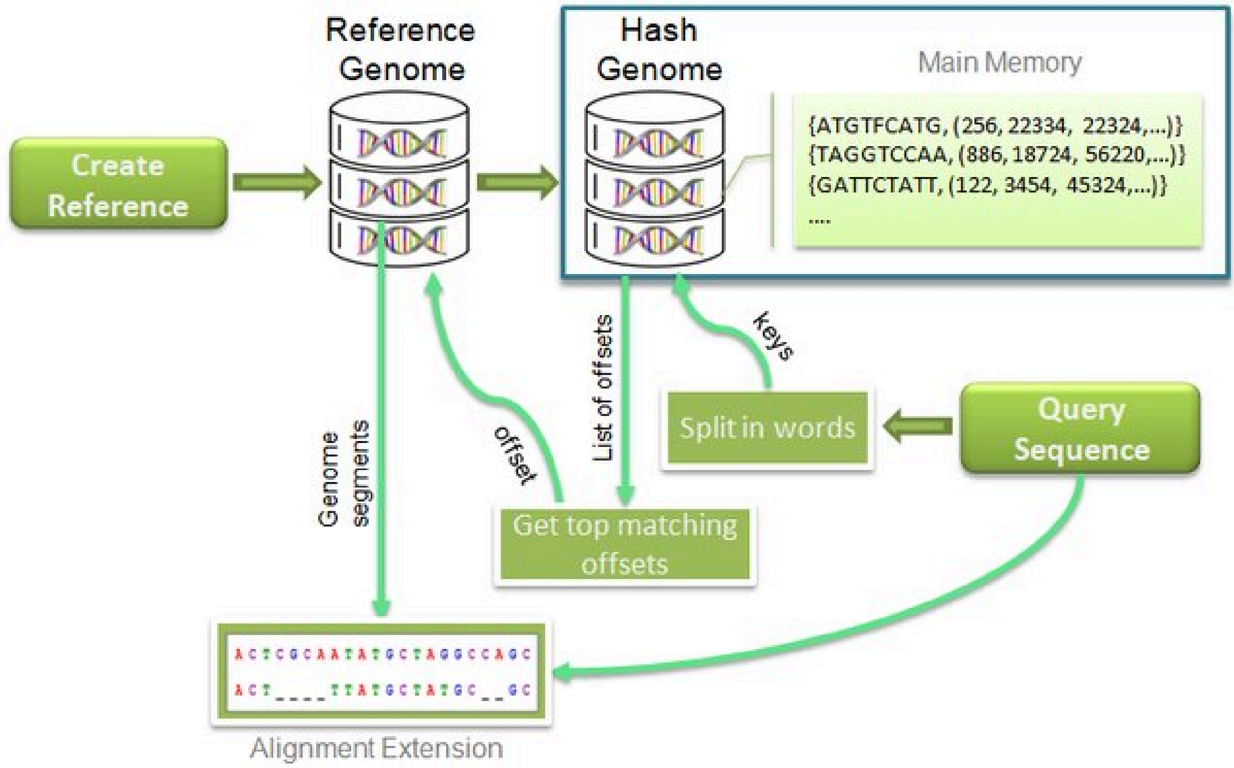
Busquen un compromís entre velocitat i precisió. No garanteixen l'òptim matemàtic, però són suficientment bons i molt més ràpids.

- Redueixen la complexitat a:

$$O(N + M)$$

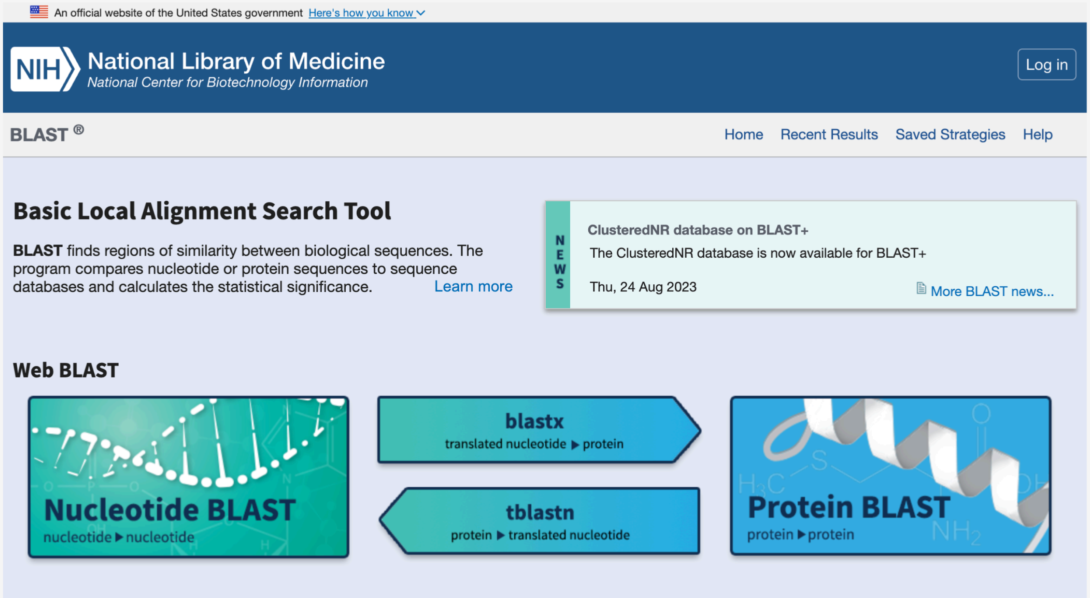
- **BLAST** (*Basic Local Alignment Search Tool*) és l'algoritme més utilitzat actualment.

https://www.researchgate.net/publication/341711005_High_throughput_BLAST_algorithm_using_spark_and_cassandra/figures?lo=1



Resultats BLAST

A GUIDE TO blast: https://chanzuckerberg.zendesk.com/hc/en-us/articles/360050963352-A-Guide-to-BLAST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Basic Local Alignment Search Tool
BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. Learn more
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Thu, 24 Aug 2023
More BLAST news...
Web BLAST
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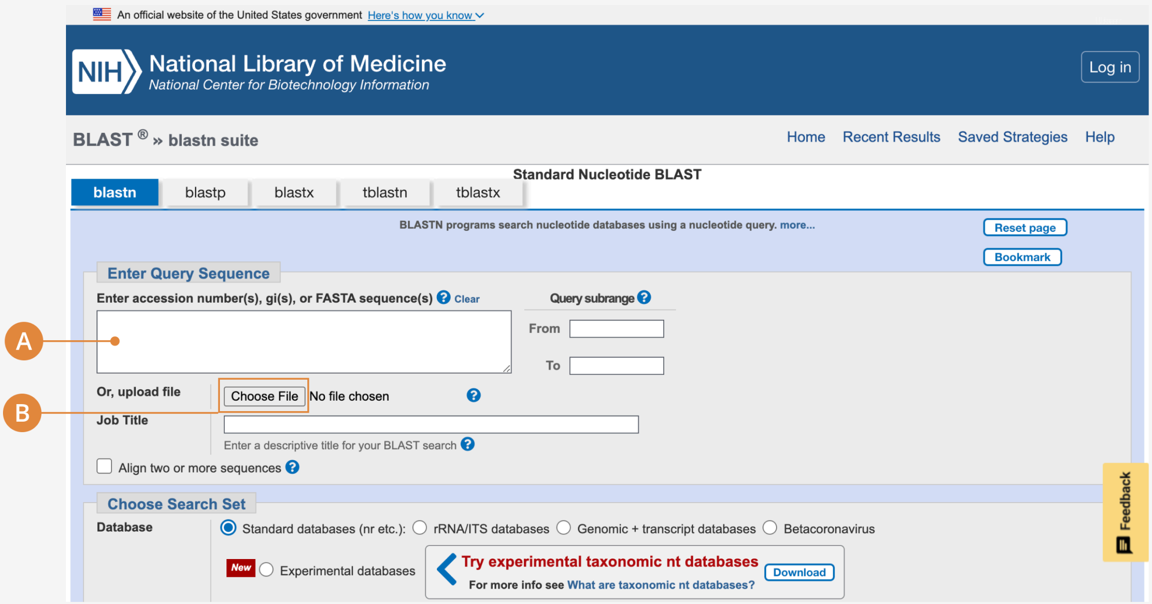
translated nucleotide ▶ protein
tblastn
protein ▶ translated nucleotide
Protein BLAST
protein ▶ protein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Job Title
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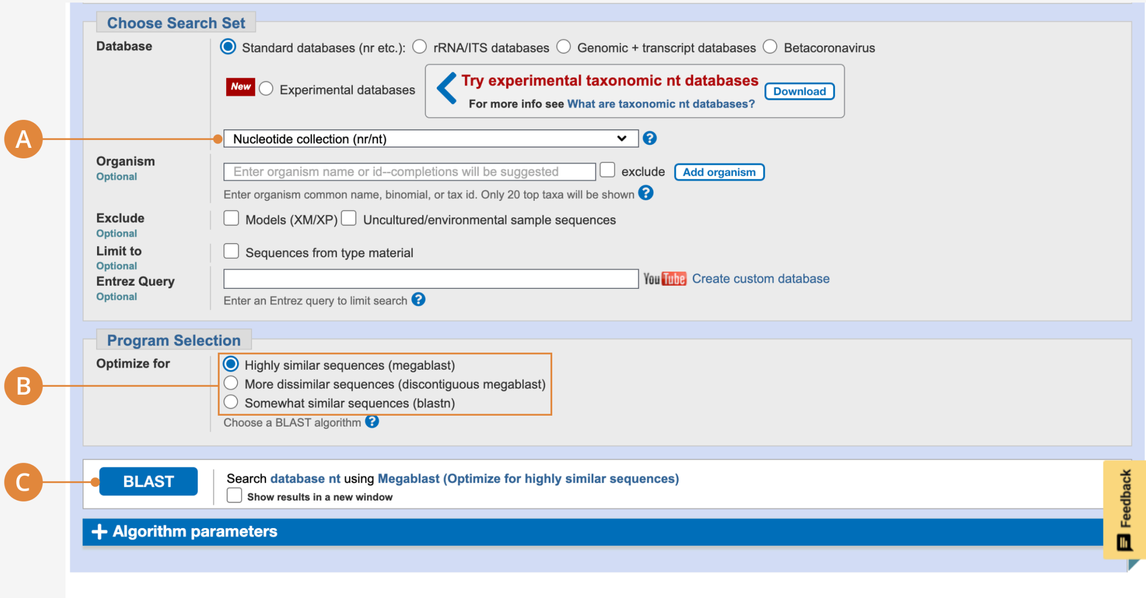
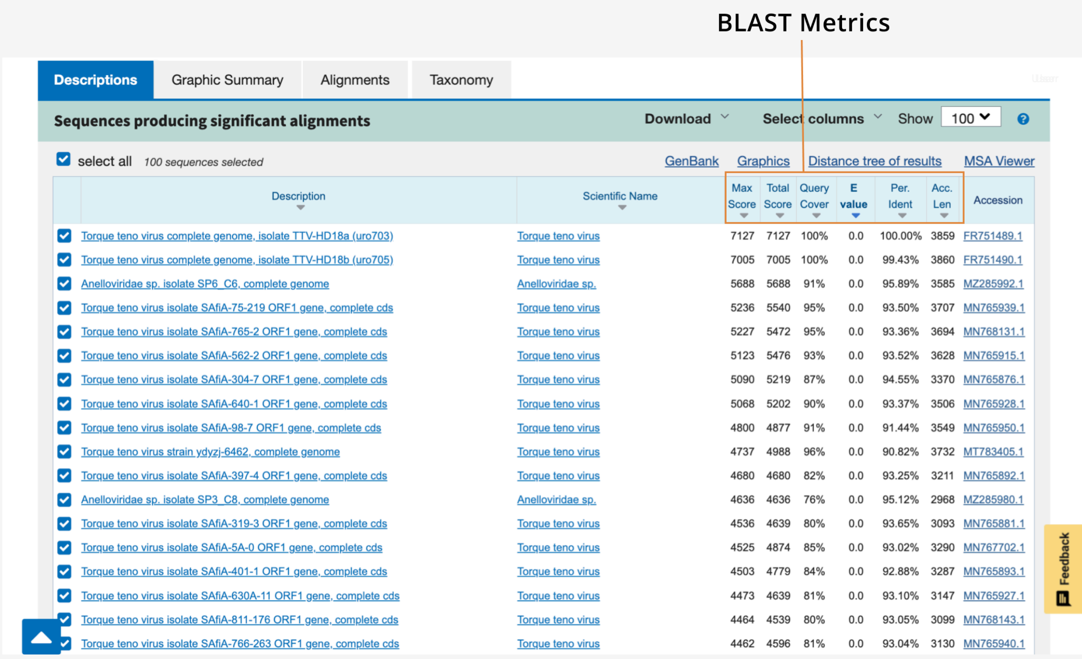
Align two or more sequences
Choose Search Set
Database
Standard databases (nr etc.): rRNA/ITS databases Genomic + transcript databases Betacoronavirus
New Experimental databases
Try experimental taxonomic nt databases
Download
For more info see What are taxonomic nt databases?
Feedback



Interpreting BLAST Results BLAST results show all of the taxa that share sequence similarity with the query sequence based on the selected database. The results page includes a search summary, hit description table, graphic summary, and alignments that can help determine the quality or accuracy of a given hit. Click here for a glossary of BLAST terms and additional information regarding alignment metrics



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Torque teno virus complete genome, isolate TTV-HD18a (uro703)	Torque teno virus	7127	7127	100%	0.0	100.00%	3859	FR751489.1
Torque teno virus complete genome, isolate TTV-HD18b (uro705)	Torque teno virus	7005	7005	100%	0.0	99.43%	3860	FR751490.1
Anelloviridae sp. isolate SP6_C6, complete genome	Anelloviridae sp.	5688	5688	91%	0.0	95.89%	3585	MZ285992.1
Torque teno virus isolate SAfiA-75-219 ORF1 gene, complete cds	Torque teno virus	5236	5540	95%	0.0	93.50%	3707	MN765939.1
Torque teno virus isolate SAfiA-765-2 ORF1 gene, complete cds	Torque teno virus	5227	5472	95%	0.0	93.36%	3694	MN768131.1
Torque teno virus isolate SAfiA-562-2 ORF1 gene, complete cds	Torque teno virus	5123	5476	93%	0.0	93.52%	3628	MN765915.1
Torque teno virus isolate SAfiA-304-7 ORF1 gene, complete cds	Torque teno virus	5090	5219	87%	0.0	94.55%	3370	MN765876.1
Torque teno virus isolate SAfiA-640-1 ORF1 gene, complete cds	Torque teno virus	5068	5202	90%	0.0	93.37%	3506	MN765928.1
Torque teno virus isolate SAfiA-98-7 ORF1 gene, complete cds	Torque teno virus	4800	4877	91%	0.0	91.44%	3549	MN765950.1
Torque teno virus strain ydyzj-6462, complete genome	Torque teno virus	4737	4988	96%	0.0	90.82%	3732	MT783405.1
Torque teno virus isolate SAfiA-397-4 ORF1 gene, complete cds	Torque teno virus	4680	4680	82%	0.0	93.25%	3211	MN765892.1
Anelloviridae sp. isolate SP3_C8, complete genome	Anelloviridae sp.	4636	4636	76%	0.0	95.12%	2968	MZ285980.1
Torque teno virus isolate SAfiA-319-3 ORF1 gene, complete cds	Torque teno virus	4536	4639	80%	0.0	93.65%	3093	MN765881.1
Torque teno virus isolate SAfiA-5A-0 ORF1 gene, complete cds	Torque teno virus	4525	4874	85%	0.0	93.02%	3290	MN767702.1
Torque teno virus isolate SAfiA-401-1 ORF1 gene, complete cds	Torque teno virus	4503	4779	84%	0.0	92.88%	3287	MN765893.1
Torque teno virus isolate SAfiA-630A-11 ORF1 gene, complete cds	Torque teno virus	4473	4639	81%	0.0	93.10%	3147	MN765927.1
Torque teno virus isolate SAfiA-811-176 ORF1 gene, complete cds	Torque teno virus	4464	4539	80%	0.0	93.05%	3099	MN768143.1
Torque teno virus isolate SAfiA-766-263 ORF1 gene, complete cds	Torque teno virus	4462	4596	81%	0.0	93.04%	3130	MN765940.1

- BLAST metrics include:
    - Max Score: Highest bit score calculated from matches and mismatches found in local alignments. The higher the max score, the better the

alignment.

  - Total Score: Sum of alignment scores for all of the sequence segments or local alignments. The higher the score, the better the alignment. When max and total scores are the same, there is one global alignment between the query and its match in the database. This means that the sequences can be aligned without long insertions or deletions. Note: CZ ID uses an algorithm equivalent to "total score" to assign taxon matches. If you find taxon A in CZ ID's Sample Report and run BLAST, it is possible that taxon A would not be the top hit in the BLAST table because BLAST sorts by E-value. Additionally, CZ ID implements a different alignment strategy (minimap2 and DIAMOND).
  - Query Coverage: Percent of the query sequence length that is included in alignments against the sequence match.
  - E-value: Indicates the number of hits or alignments that are expected to be seen by random chance with the same score or better. The lower the E-value, the more significant the alignment (the closer to 0, the better). E-value is the default metric used to sort the Descriptions table. Click here for a discussion of E-value thresholds.
  - Percent Identity: Percent of nucleotides or amino acids that are identical between the aligned query and database sequences. A query sequence can share low percent identity with a sequence and still be a significant hit. It is essential to take the E-value into account and look for similarity between conserved regions (this will be more evident at the amino acid level).

MES RESULTATS I EXPLICACIONS A: https://chanzuckerberg.zendesk.com/hc/en-us/articles/360050963352-A-Guide-to-BLAST

Interpretació de resultats: El E-value

El paràmetre més important per filtrar resultats no és només el % d'identitat, sinó el **E-value (Expect value)**.

- **Definició:** Expressa el nombre d'alineaments esperats amb una puntuació igual o superior a l'obtinguda si la similitud fos deguda purament a l'atzar.
- **Interpretació:** És una esperança, no una probabilitat. Quan més petit és el valor, menys probable és que el resultat sigui per atzar.

La fórmula aproximada és:

$$E = K \cdot m \cdot n \cdot e^{-\lambda \cdot S}$$

On:

- m, n: Longitud de la seqüència problema i mida de la base de dades.
- S: Puntuació (score).
- K, λ: Factors d'escala.

Criteris de Consens

Per considerar una possible homologia:

1. **Esperança:** $E() < 0.05$
2. **Identitat:** $> 20 - 25\%$
3. **Context:** Informació biològica (funció/estructura).

Instruccions Generals

- **Eina:** Les tasques següents les farem amb l'opció **'Protein BLAST' (`blastp`)**, ja que volem fer cerques de similitud de seqüències proteiques en bases de dades de proteïnes.
- **Base de dades:** Utilitzarem **'Non-redundant protein sequences (nr)'**.
- **Objectiu:** Intenteu respondre a les preguntes plantejades. Quan parlem de si són *estadísticament significatius*, ens referim a que interpreteu els valors de $E()$ (*E value*) que obteniu, però recordeu que també són importants els valors de `Per. Ident`, els `scores` i `Query Cover`.

> **Nota:** No farem una explicació detallada prèvia dels resultats. Es tracta que, aplicant la informació de la teoria, intenteu pensar la interpretació per vosaltres mateixos.